Priyanshu Ranjan

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Professional Summary

Computer Science undergraduate specializing in AI & Machine Learning with a CGPA of **8.76/10**. Experienced in exploratory data analysis, model evaluation, data warehousing, and BI dashboards using Python, SQL, Pandas, NumPy, Power BI, and Tableau. Built analytics solutions across Aadhaar, customer churn, and retail transaction datasets to identify trends, monitor KPIs, and support data-informed decisions.

Education

Vellore Institute of Technology **Sep 2023 – Expected 2027**
B.Tech in Computer Science & Engineering (AI & ML Specialization) | ***CGPA: 8.76/10***

Projects

Aadhaar Insights - Data Analytics Platform **Jun – Jul 2026**

Large-Scale Data Analytics — Python, Pandas, NumPy, Scikit-learn, Streamlit, Power BI, Plotly

- Built a Python analytics platform processing **3.37M+ Aadhaar records** across **3 official UIDAI datasets** to examine enrolment, demographic, and biometric trends.
- Developed a **9-page Streamlit application** and Power BI dashboards for KPI monitoring, anomaly detection, trend and correlation analysis, and automated recommendations.
- Compared **3 regression models**—Baseline, Linear Regression, and Random Forest—across the three datasets using MAE, RMSE, MAPE, and R^2 metrics.

Customer Churn Prediction - Analytics Platform **Aug – Sept 2026**

Customer Analytics & Explainable ML — Python, Pandas, Scikit-learn, XGBoost, SHAP, Tableau

- Analyzed **7,043 Telco customers**; on the project's **1,407-customer test split**, month-to-month churn was **41.3%** versus **2.4%** for two-year contracts.
- Built a Tableau dashboard for churn and retention KPIs; on the same test split, churn was **44.8%** for electronic-check customers versus **14.7%** for credit-card customers.
- Benchmarked **5 classifiers** with stratified evaluation and tuned XGBoost using **5-fold cross-validation**; the repository reports **84.81% ROC-AUC** and **60.06% F1** on its test split.

Smart Retail Analytics & Recommendation Warehouse **Feb – Apr 2026**

Retail BI & Customer Segmentation — SQL, SQLite, Pandas, Scikit-learn, Apriori

- Built an end-to-end retail analytics pipeline on the UCI Online Retail dataset (approximately **541K transactions**), transforming cleaned records into a SQLite star schema with **392K fact rows** and **4 dimension tables**.
- Implemented **15+ OLAP queries** covering drill-down, roll-up, slice, and dice analysis to explore transaction and product trends.
- The team project uses RFM scoring and K-Means to segment **4,338 customers into 4 clusters**; its market-basket analysis produced **68 association rules**, with a maximum lift of **24.0**.

Technical Skills

Languages	Python, SQL, Java
Data Analysis	Pandas, NumPy, EDA, Data Preprocessing
BI / Visualization	Power BI, Tableau, Streamlit, Plotly
Analytics / ML	Scikit-learn, Regression, Classification, Model Evaluation
Warehouse / Methods	SQLite, OLAP, RFM, Association Rules
Tools	Git, Google Colab

Certifications & Achievements

- **Oracle Cloud Infrastructure 2025 Certified Data Science Professional** — Oracle
- **Oracle Agentic AI Foundations Associate (1Z0-1157-26)** — Oracle
- **Applied Machine Learning in Python** — Certificate
- **TCS CodeVita Season 13 – Global Rank: 1206** — Proof
- **Graph Camp – AlgoUniversity** — Certificate